Supplementary Material for WEAVE

Anonymous submission

Abstract

This supplement provides the implementation, evaluation, and diagnostic details needed to reproduce the main paper. Paths are package-relative. The analytic fiber scale β is an inference-time solver parameter; main-table evaluation uses `inference.json` with β=2.0.

Reproducibility Overview

The anonymous archive is a **minimal code package**: source, configs, and the selected D5 checkpoint. Datasets, packed latents, and ONNX/TensorRT graphs are **not** redistributed in the archive (size limits); they are prepared externally following this supplement.

Included in the anonymous archive.

Path	Role
`configs/hf_oriented_internal_early_stop.json`	D5 train config (internal early stop).
`config.json/inference.json`	Base config; main-table eval override (β=2.0).
`checkpoints/weave_epoch_0004.pt`	Selected D5 checkpoint.
`tools/prepare_submission_data.py`	Path sanitize for local data layout.
`docs/generation_only_timing_summary.json`	Audited 73.0 s / 750 timing ledger.
`docs/onnx_export_and_trt.md`	How to export ONNX and compile TensorRT.
`data/README.md`	Expected local data layout (not the data itself).

Required externally (not in the minimal archive).

Path	Role
`data/train`, `data/test`	D5 latents + eval images (build from WikiArt).
`data/train_p2a`, `data/test_p2a`, `data/images/r5`	P2A/R5 boards if those rows are re-run.
`data/onnx_exported/` (optional)	VAE ONNX graphs for the 73 s timing protocol.
Collection-specific P2A/R5 checkpoints	Needed to reproduce those Table 1 rows.

The shipped D5 checkpoint alone does **not** reproduce the collection-specific P2A or R5 rows in main-paper Table 1.

Datasets and Evaluation Boards

D5-512. The primary benchmark is Distinct5-WikiArt at 512 px. It contains five WikiArt styles: Early Renaissance, Impressionism, Minimalism, Rococo, and Ukiyo-e. Training uses 5,000 images (1,000 per family; the source pool is larger, but a fixed 1,000-image subset per family is used, matching the 52 steps/epoch at batch 96). Evaluation uses all ordered source-target style pairs with 30 content images per source style, giving 750 pairs. The 150 same-style pairs are retained as identity calibration rows rather than removed.

P2A-256. The photo-to-art board evaluates transfer at 256 px. It uses the same ordered evaluation logic but with a lower-resolution latent/image cache. It is included to test whether the method remains stable when high-frequency subbands are compressed by resolution.

R5-WikiArt. R5 tests collection-specific training on five WikiArt styles drawn at random from WikiArt and held out from D5: Cubism, Expressionism, Pop Art, Romanticism, and Symbolism (`data/images/r5/test`; 30 images per style). Evaluation uses the same ordered 750-pair protocol as D5 ($5 \times 5 \times 30$). Other5 (main-paper zero-shot table) is a different five-family list held out from D5; it is not required to be disjoint from R5.

Pairing and calibration. All learned methods use the same ordered pair lists within a board. For D5 training, source-target pairing uses `data/train/.latent_cache/prototype_pairing_top8.pt` with `pairing_cache_active_topk=0`. Identity rows estimate the no-op style floor and content ceiling.

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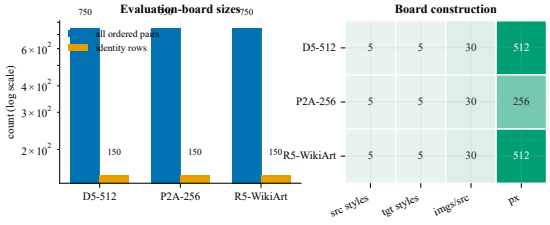


Figure S1: Evaluation-board construction. D5-512, P2A-256, and R5-WikiArt each use a 750-pair board with identity rows retained. R5 is five randomly selected held-out WikiArt styles (Cubism, Expressionism, Pop Art, Romanticism, Symbolism).

Dataset Layout

Expected local layout after preparing WikiArt boards (see `data/README.md` in the code package; data themselves are external):

Path	Contents
`data/train`	D5 latents + `.latent_cache/packed` (train default).
`data/test`	D5-512 held-out eval images (150).
`data/train_p2a`, `data/test_p2a`	P2A latents and eval images.
`data/images/d5/train`	D5 train JPGs (optional re-encode).
`data/images/r5/\{train,test\}`	R5 WikiArt board.
`data/onnx_exported/`	Optional VAE ONNX graphs (export; not shipped).

Expected counts. D5/R5 boards: $5{\times}5{\times}30{=}750$ ordered pairs (identity rows retained). P2A uses the same ordered-pair protocol at 256 px; some external baselines provide only the 600 off-diagonal transfers.

Data Preprocessing Contract

Training consumes latents (VAE scale 0.18215), not raw RGB. Style directory names must match the config lists. Pairing caches and metric reference caches are built only from training latents and held-out test/style-bank images, never from generated outputs.

Hardware and Software Environment

All training and timing numbers reported in the paper come from a single consumer-class GPU. The environment is described by role, not by machine identifier:

Component	Value
GPU class	NVIDIA GeForce RTX 3060, 12 GB
Training precision	bfloat16 automatic mixed precision
TF32 matmul	enabled
Device count	1 GPU

The method itself does not depend on a specific software stack version beyond the ranges listed above; any consumer GPU with ≥ 12 GB VRAM and a working PyTorch CUDA build reproduces the reported numbers.

The software stack follows `WEAVE/requirements.txt`: PyTorch $\geq$ 2.3, torchvision $\geq$ 0.18, transformers $4.57 \leq v < 5$, diffusers $0.38 \leq v < 0.39$, LPIPS, torch-metrics, torch-fidelity, Pillow, NumPy, SciPy, scikit-learn, and tqdm. The frozen VAE is the SD1.5 EMA VAE used only for latent encoding/decoding; it contributes 84M frozen parameters and is not trained.

Model and Training Setup

WEAVE trains a compact velocity model in SD1.5 VAE latent space. A single-level Haar DWT splits the latent into LL/LH/HL/HH subbands. The base network has four residual blocks of width 64 and three supervised output heads for LL/LH/HL. The D5 model adds separate pooled target-LH and target-HL encoders whose gated residuals enter only the matching velocity heads. The complete model has 1,037,087 trainable parameters; the frozen VAE is reported separately as $1.04\text{M}^{+84\text{M}}$.

Field	Paper configuration
Config	`configs/hf_oriented_internal_early_stop.json`
Base config	`config.json`
Seed	42
Maximum epochs	15; internal stop at epoch 4
Batch size	96
Gradient accumulation	1
Optimizer	AdamW, fused when available
Learning rate	2.0×10^{-4}
Minimum learning rate	1.0×10^{-5}
Schedule	cosine
Weight decay	1.0×10^{-4}
Gradient clipping	1.0
Precision	bf16 AMP
Channels-last layout	enabled
Torch compile	disabled
Workers	0
Checkpoint interval	every epoch
Internal probe batch	4 fixed latents at (t=0.5)

The structure-aligned target is constructed in the wavelet basis. The LL target is a mild content-preserving blend,

$$\begin{aligned}\mathrm{LL}_{target} = {} & 0.7\,\mathrm{LL}_{content} \\ & + 0.3\,\mathrm{AdaIN}(\mathrm{LL}_{content}, \mathrm{LL}_{style}),\end{aligned}$$

while the high-frequency target uses the target-style LH/HL/HH bands. The implementation minimizes the band-weighted velocity-matching loss given in the main paper; no additional edge or low-pass loss is active in the submitted configuration.

Resolved Configuration Details

The frozen merged configuration for the paper run resolves the short public config into the following relevant fields:

Field	Value	Relevance
model.contract_family	weave	Selects the WEAVE architecture contract.
model.base_dim	64	Backbone width.
model.time_dim	256	Time embedding width.
model.num_res_blocks	4	Main residual block count.
model.num_hires_blocks	2	High-resolution block count.
model.num_decoder_blocks	2	Decoder block count.
model.target_latent_hf_subband_fusion_enabled	true	Activates orientation-specific target-HF residuals.
model.target_latent_hf_head_fusion_gate_init	0.18	Initial LH/HL route gate.
model.lowpass_mode	dwt_haar	Single-level Haar decomposition.
model.endpoint_adain_mode	spatial_fiber	High-frequency residual statistics matched after every Euler step.
model.endpoint_adain_only_last_step	false	Keeps AdaIN inside all eight integration updates.
model.endpoint_adain_scale	base 1.0; reported eval 2.0	$= \beta$ in the main-paper fiber update S_s. β affects solver evaluation only and does not enter Eq. of the main-paper training loss. The base checkpoint stores 1.0; main Table 1 uses the inference.json override 2.0, which recovers the reported D5 board means.
model.style_extrap_alpha	0.1	$= \eta$: multiplies target HF statistics before matching.
bridge.objective_mode	flow_matching	Noise-regularized endpoint-velocity regression.
bridge.loss_type	mse	Velocity matching loss.
training path	$t \sim \mathcal{U}[0, 1]$	Time sampling used by the submitted flow objective.
bridge.bridge_sigma	0.02	Scales the $\sqrt{t(1-t)}$ interpolation perturbation.
bridge.training_target_projection_mode	dwt	Target construction in wavelet space.
bridge.spectral_w_ll/lh/hl/hh	0.3/1.0/1.0/inactive	HH has no velocity head in WEAVE.
bridge.ll_partial_alpha	0.3	Mild LL style-statistics migration.
training.internal_probe_enabled	true	Measures fixed-batch loss-path gradients after each epoch.
training.internal_early_stop_enabled	true	Stops at the first gate/gradient transition.

Table S1: Resolved fields from the frozen paper configuration.

Training Log Audit

The D5 paper run contains four logged epochs, each with 52 optimizer steps and effective batch size 96. The wall-clock epoch time sums to 82.80 s. The probe is evaluated after the optimizer and scheduler step, before checkpoint saving, and does not decode images.

Epoch	Loss	Learning rate	Epoch time (s)	Stop event
1	1.3631	2.0000e-4	25.72	no
2	1.3318	1.9792e-4	19.23	no
3	1.3199	1.9179e-4	18.94	no
4	1.2499	1.8186e-4	18.90	yes

The first epoch is slower because caches and kernels are warmed. Later epochs take about 19 s on a single RTX 3060 (12 GB), including the fixed-latent probe.

Internal-Dynamics Probe

For a fixed latent minibatch, the implementation performs one backward pass for the LL loss and one for the summed LH/HL loss. Gradients are aggregated over the shared input, attention, AdaLN, and feed-forward modules. The target-HF route and ordinary HF heads are measured separately. The model runs in evaluation mode during the probe, no optimizer step is taken, and torch.random.fork_rng restores CPU and CUDA random states.

Epoch	Gate	ΔGate	Shared LL/HF	Route/shared	Route/head
1	0.173068	0.000000	1.4727	0.8323	0.1816
2	0.172264	-0.000803	1.4328	0.6133	0.1907
3	0.172009	-0.000255	1.0868	0.4984	0.2111
4	0.175719	+0.003710	0.1634	0.2370	0.4561

Table S2: Stopping-trajectory probe on the D5-512 checkpoint (one fixed latent minibatch per epoch).

Epoch 4 is the first joint event where the gate derivative is positive and the shared LL/HF ratio contracts by at least 35% from the preceding epoch. A retrospective probe of the complete 15-epoch curve selects the same epoch, whose DINO-S is the curve maximum. Residual alignment to the FM target, however, continues to improve after epoch 4 while DINO-S declines. This separates the proposed transition from ordinary loss-based early stopping: it detects a change in route ownership rather than a plateau in endpoint regression.

The original absolute crossing-at-one rule was scale-sensitive. Replacing it by $\rho_e/(\rho_{e-1} + \epsilon) \leq 0.65$ selects epoch 4 for probe batches 2/4/8 and seeds 42/7, and epoch 3 for seed 123. The additional seed-7 and seed-123 checkpoints give (DINO-

Algorithm S1 Evaluation-free internal-dynamics stopping rule

Require: trained WEAVE trunk; probe minibatch $\mathcal{B}$; ratio threshold τ=0.65; expansion tolerance ϵ>0

run one LL-loss and one summed LH/HL-loss backward pass on $\mathcal{B}$ (eval mode, no optimizer step)
record target-HF gate g_e and shared-trunk LL/HF gradient norm ρ_e
for $e = 1, 2, \ldots$ **do**
    compute $\Delta g_e = g_e - g_{e-1}$ and ratio $r_e = \rho_e/(\rho_{e-1} + \epsilon)$
    if $\Delta g_e > 0$ **and** $r_e \leq \tau$ **then**
        return selected epoch e ▷ first joint gate-expansion & $\geq 35\%$ LL/HF contraction
    end if
end for

Ensure: never generate images; never consult DINO-S, CLIP-S, LPIPS, or DINO-C

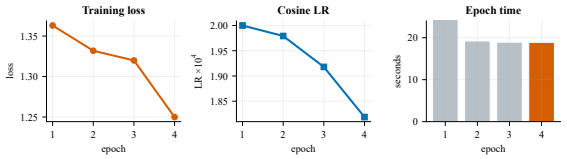


Figure S2: Training dynamics for the selected D5-512 checkpoint. Training stops at epoch 4 when the high-frequency conditioning gates increase and the shared LL/HF gradient ratio contracts sharply relative to the preceding epoch.

S, CLIP-S, LPIPS, DINO-C) of (0.4910, 0.7140, 0.2668, 0.8076) and (0.4862, 0.7144, 0.2552, 0.8040), respectively. Thus the event is stable to the tested probe scale and keeps the selected point near the canonical frontier; it is not claimed as an architecture-independent criterion.

Implementation Notes

Core modules: `run.py`, `trainer.py`, `internal_dynamics.py`, `model.py`, `wavelet.py`, `utils/run_evaluation.py`. Material details: VAE scale `0.18215`; training supervises LL/LH/HL heads (HH via analytic alignment); evaluation metrics use held-out style-bank caches only.

Information-Flow Diagnosis

Closed-form Pixel–Latent Representability

We test whether appearance changes that are simple in RGB remain simple after VAE encoding, without training a network. The test uses 96 source-disjoint requests at 256×256 resolution. For each request, we construct four RGB targets: a channel-wise color shift, AdaIN, WCT, and a channel-wise gamma curve. The target latent is the deterministic posterior-mean encoding of that same transformed image; posterior sampling is not used.

In each representation, we independently solve the optimal cross-channel 1×1 affine map by least squares over all spatial positions of each request. We report its squared endpoint residual divided by the true endpoint-displacement energy. This normalization makes zero displacement equal to one in either representation and avoids comparing raw RGB and latent MSE scales.

RGB target transformation	Pixel NMSE	VAE-latent NMSE
Color shift	0.0072	0.2527
AdaIN	0.0122	0.3138
WCT	0.0143	0.3765
Gamma curve	0.0411	0.2937

The small nonzero RGB residuals arise from clipping and, for the gamma target, from its nonlinearity. The much larger latent residual remains even though the fit is an optimistic per-request oracle. The result therefore isolates a representation effect: VAE compression reduces spatial computation, but a pointwise RGB appearance change need not remain pointwise or affine in latent coordinates.

As a matched nonlinear check, we train the same four-layer 1×1 conditional MLP in both representations for 1,500 optimizer steps and three seeds. Both versions use width 64, SiLU activations, identical request batches and sampled spatial positions, and the same RGB content/style means and standard deviations as conditioning. This is a separate pointwise representation diagnostic, not an image-quality evaluation.

Representation	Endpoint NMSE
Pixel	0.0327 ± 0.0004
VAE latent	0.3126 ± 0.0041

The latent curve plateaus near its closed-form affine residual, whereas Pixel continues toward the target.

Main-paper Figure 4(b) uses a spatial TinyUNet with approximately 7.6M parameters, 480 training pairs, 30 epochs, and three seeds. All variants use the same global velocity MSE. Pixel and plain latent transport use the RGB-AdaIN endpoint in their respective representations; the third variant changes only the latent endpoint to LL from the source and LH/HL/HH from the style.

Transport setting	Final NMSE	Seeds
Pixel, RGB-AdaIN endpoint	0.0418 ± 0.0029	3
VAE latent, RGB-AdaIN endpoint	0.3542 ± 0.0060	3
VAE latent, spectral endpoint	0.0086 ± 0.0004	3

The frequency-decoupled endpoint is substantially easier for the same network to regress. This is an internal learnability result, not an external image-quality comparison.

Latent Subband Statistics

We apply one-level orthonormal Haar decomposition to 5,000 deterministic SD1.5 EMA-VAE latents from five Distinct5 styles. For each latent, subband energy is normalized by total latent energy. Style separability uses 500 latents (100 per style): the scalar response is the per-image mean absolute coefficient in each band, and the reported η^2 is the one-way ANOVA effect size of the style label.

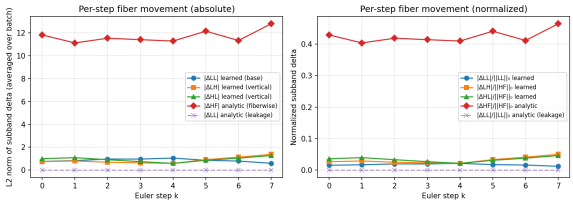


Figure S3: Per-step routed motion over eight cross-style probes. S_s supplies the dominant HF update, while analytic LL leakage remains below 3.2×10^{-6}.

Band	Mean energy	p5	p95	Style η^2
LL	0.659	0.459	0.904	0.436
LH	0.115	0.032	0.184	0.270
HL	0.125	0.038	0.201	0.375
HH	0.101	0.021	0.171	0.499

Because Haar is orthonormal, the energy shares are directly comparable and sum to one. They show a clear scale imbalance rather than an extreme low-pass collapse. The ANOVA result also rejects a hard semantic assignment of LL to content and HH to style: every band is style-sensitive, with a non-monotonic orientation pattern and the largest effect in HH. These measurements motivate separate frequency routes and partial LL movement; they do not determine the numerical LL blend or loss weight, which are evaluated by ablation.

Terminal frequency dynamics. We also evaluate matched 10-epoch checkpoints with $\lambda_{LL} = 0.3$ and 1.0 on the same fixed batch of four cross-style pairs at $t = 0.5$ (seed 42). The values below are a terminal internal snapshot, not an average over training.

λ_{LL}	LL loss	HF loss	$\|g_{LL}\|_2$	$\|g_{HF}\|_2$
0.3	3.03×10^{-2}	1.144	0.592	1.611
1.0	2.94×10^{-8}	1.144	4.27×10^{-7}	2.113

Uniform weighting reaches a nearly zero LL residual and LL gradient while its HF state is essentially unchanged. The weighted model retains active gradients in both routes. This supports monitoring and weighting bands separately, but a single terminal batch does not by itself identify parameter capacity as the cause.

Boundary of the HH-path diagnostic. Across 50 cross-style pairs, the directly comparable output correlations are 0.0245 ± 0.0108 for an HH-head reconstruction and 0.0238 ± 0.0115 for AdaIN; magnitude correlations are respectively 0.0316 ± 0.0119 and 0.0294 ± 0.0137. Thus this test finds no meaningful increase in output-level target topology from the HH head. Its velocity correlates with the supervised HH displacement (0.2086 ± 0.1275), but AdaIN has no corresponding velocity variable, so these quantities should not be divided into a leakage factor. We therefore treat the HH-head choice as the style–content trade-off reported by the ablation, not as evidence of inevitable geometric leakage.

The probe campaign was designed to answer a narrower question than the main table: whether the model's weak style movement came from a wrong target, insufficient training time, or a disconnected conditioning path. The diagnosis favors the third explanation. The structure-aligned target already asks for style in the high-frequency bands:

$$\begin{aligned} \mathrm{LL}_{target} &= 0.7\,\mathrm{LL}_{content} \\ &\quad + 0.3\,\mathrm{AdaIN}(\mathrm{LL}_{content}, \mathrm{LL}_{style}), \\ \mathrm{LH/HL/HH}_{target} &= \mathrm{LH/HL/HH}_{style}. \end{aligned}$$

Thus the loss is not merely reconstructing the source image. The failure mode is that, in the baseline, the target image latent is used to construct supervision but does not strongly enter the velocity predictor as an image-specific condition.

Route	Signal path	Probe reading
Baseline style memory	`style_id` → style memory → cross-attention	Coarse style prior. Gates stay near 0.056–0.061, but removing this route hurts full metrics, so it is useful but not sufficient.
Global target-token fusion	target latent → all style tokens → backbone	Connected but too global; it over-controls LL and does not improve useful style transfer.
Raw spatial target-HF	target DWT-HF maps → HF residual maps	Has sufficient capacity, but leaks target geometry and collapses content.
Pooled subband target-HF	target DWT-HF → per-subband pooled code → HF residual velocity	Current usable path: stronger style signal without target-coordinate leakage.

The strongest negative control is the raw spatial target-HF route. It raises DINO-S and CLIP-S, which shows that the network can read target-style information when the path is opened. However, the same run reduces DINO-C to 0.4043 and raises LPIPS to 0.5382, so the gain is not a valid style-transfer improvement. It is target-layout leakage. The accepted route therefore compresses target high frequencies into coordinate-free subband codes before they reach the HF velocity heads.

Probe Protocol and Result Ledger

Diagnostic target-HF probes fine-tune from a shared base checkpoint and evaluate on D5. The main accepted lessons are:

- Spatial target-HF injection can raise style scores but leaks target geometry (DINO-C collapses; LPIPS rises sharply) and is rejected.
- Coordinate-free pooled/subband target-HF codes preserve content while retaining most of the style gain; this is the route used by the paper D5 model.
- Zeroing subband residuals at inference lowers both style and content metrics, so the learned residual path is active rather than dead weight.
- Scalar residual amplification, time-windowing, FiLM heads, and memory-prior subtraction do not improve the balanced operating point over the simple subband code.

- Style memory is a useful coarse prior: removing it weakens transfer, but it cannot replace image-specific HF codes.

Representative board means (DINO-S / DINO-C / LPIPS / CLIP-S): spatial 0.490/0.404/0.538/0.748 (reject); subband 0.489/0.798/0.297/0.721 (accepted family); residual ablated 0.486/0.789/0.301/0.721.

Inference Setup

The main paper uses an 8-step Euler solver. Spatial-fiber AdaIN is applied to the high-frequency residual after every solver step. The main D5-512 row uses `runs/submission/canonical_oriented_epoch4/epoch_0004.pt` (oriented target-HF route). P2A and R5 use audited base-WEAVE checkpoints with the same Haar endpoint and stepwise-alignment formulation but without the later oriented D5 route.

Inference field	Value
ODE solver	Euler
ODE steps	8
Generation batch size	8 (ONNX/TRT pipeline; encoder+WEAVE fused stage)
VAE encode backend	ONNX Runtime FP16 (`encoder_b8.onnx`)
VAE decode backend	TensorRT FP16, workspace 2 GB (`decoder_b8.onnx`)
Spatial-fiber AdaIN	every Euler step; main-table eval uses `endpoint_adain_scale` = 2.0 via `inference.json` (β in the main paper; train uses 1.0)
Style extrapolation	η = `style_extrap_alpha` = 0.1
Generated-image count for timing	750
Timing column	end-to-end generation wall clock (encode + ODE + decode; metrics excluded)

Evaluation Runtime Anatomy

The paper reports the measured generation-only time for the selected D5 checkpoint: 73.0 s for 750 images (97.3 ms/image) on one RTX 3060 (12 GB) under the audited ONNX/TRT pipeline (batch 8, decoder workspace 2 GB). Stage breakdown: encoder+WEAVE 14.0 s (18.7 ms/image, 19%), TRT decoder 59.0 s (78.6 ms/image, 81%). The timing includes the 8-step trajectory and VAE encode/decode, but excludes latent caching, metric extraction, and result serialization. Larger decode batch (16) triggers TRT build OOM on a 12 GB card and is not used. Increasing decoder workspace from 2 GB to 4 GB improves decode by only $\approx 0.6\%$ and is not required. The latent cache itself is a one-time offline VAE encode of the full D5 training set (5,000 images at 512^2); with our fixed encoder-only ONNX graph under TensorRT ($\approx$37 ms/image) this takes about 3 minutes, and is amortized across all runs that reuse the same packed cache.

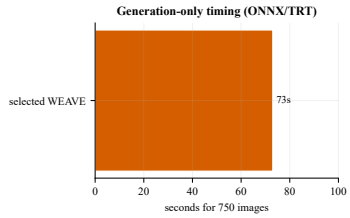


Figure S4: Generation-only time for the selected WEAVE checkpoint on a single RTX 3060 (12 GB): 73.0 s for 750 images (97.3 ms/image; ONNX/TRT pipeline). Latent caching and metric computation are excluded.

Inference Variants

The following older base-WEAVE variants share one checkpoint and differ only in inference-time stepwise AdaIN strength. They remain diagnostic operating points and are not the updated main D5 result.

The audited generation-only timing for the selected checkpoint on a single RTX 3060 (12 GB; batch 8; TRT decoder workspace 2 GB) is:

Timing field	Value
Total generation-only wall clock	73.0 s
Encoder+WEAVE stage	14.0 s (18.7 ms/img)
TRT decoder stage	59.0 s (78.6 ms/img)
Images	750
Wall time per image	97.3 ms
Throughput	10.3 pairs/s

Metrics

DINO-S. DINO-S is the primary style metric. It is the cosine similarity between the DINOv2-small CLS feature of the generated image and the nearest reference feature in the target-style reference pool. The source image is excluded from the reference pool when applicable.

CLIP-S. CLIP-S is the secondary style metric, computed as CLIP ViT-B/32 image-feature cosine similarity to the target-style reference prototype.

DINO-C. DINO-C measures content preservation as DINOv2-small CLS cosine similarity between the generated image and the source image.

LPIPS. LPIPS uses the AlexNet backbone and is reported as a lower-is-better content distance. Radar plots display $1 - \mathrm{LPIPS}$ so all axes point outward for better values.

IDT–TGT Evaluation Guideline

The guideline adds two source-independent reference outputs to the ordinary metric table. For a request with source image x and target style s, define

$$y_{\mathrm{IDT}}(x, s) = x, \qquad y_{\mathrm{TGT}}(x, s) = r_s, \tag{1}$$

where r_s is a fixed image from the held-out reference pool for style s. The D5 implementation selects the first image in the deterministic reference order and reuses it for every

Variant	Scale	DINO-S	CLIP-S	LPIPS	Use
WEAVE-m	1.5	0.4843	0.7180	0.2925	Balanced diagnostic point.
WEAVE-s	1.6	0.4845	0.7179	0.2867	Peak average of DINO-S and CLIP-S.
WEAVE-q	2.0	0.4859	0.7075	0.2583	Previous base-WEAVE max-DINO diagnostic.

Table S3: Alternative inference settings evaluated from the same base checkpoint.

source paired with that target style. Thus TGT depends on the requested style but not on source content. It is an evaluation control, not an input to WEAVE or any baseline.

Why two references are needed. IDT represents a no-op: it preserves all source content but makes no requested edit. A method should therefore score above IDT on DINO-S and CLIP-S before a low LPIPS can be read as successful transfer. TGT represents the opposite extreme: it has the requested style but unrelated content. A content-preserving method should have lower LPIPS and higher DINO-C than TGT. The intuitive acceptable region is therefore bracketed by

$$\begin{aligned} M_{\text{style}}(y) &> M_{\text{style}}(y_{\text{IDT}}), \\ \text{LPIPS}(y, x) &< \text{LPIPS}(y_{\text{TGT}}, x), \\ \text{DINO-C}(y, x) &> \text{DINO-C}(y_{\text{TGT}}, x). \end{aligned} \tag{2}$$

We inspect the inequalities separately and do not combine them into a weighted score.

Interpretation and scope. IDT and TGT are empirical anchors, not mathematical bounds: a generated image may score outside either reference because the metrics are imperfect. Crossing IDT in the wrong direction indicates style suppression under that metric. Moving beyond TGT in content distance identifies a content-loss regime relative to a deliberately unrelated target-style image. This is a prompt for visual inspection, not a verdict on the method outside the current board and metric. On D5-512, IDT is (DINO-S 0.419, CLIP-S 0.640, LPIPS 0, DINO-C 1), while TGT is (1.000, 0.855, 0.756, 0.220). TGT reaches DINO-S 1.000 because the selected reference is part of the same target-style feature pool used by that metric; this value is expected and should not be read as model performance.
Using each of the first five deterministic target exemplars changes TGT LPIPS from 0.752 to 0.783 and DINO-C from 0.189 to 0.246. StyleAligned remains beyond every TGT LPIPS anchor (0.869), so the content-collapse diagnosis is not determined by the first exemplar.

Metric Cache and Determinism

Evaluation uses deterministic seeds and cached reference features. Cache entries are logical artifacts under `<CACHE_ROOT>/eval_cache`; they may be rebuilt as long as the image roots and model backbones are unchanged. The full-eval summaries record:

- CLIP backend: HuggingFace transformers backend.
- LPIPS network: AlexNet.
- Reference-cache limit per style: 16 for the paper full-eval records.
- Source-feature cache: one cache per board/image-size setting.
- Metric postprocessing: disabled for the main records.

DINO Sidecar Coverage

The DINO sidecar contains 37 board-method entries. D5-512 and P2A-256 each include 13 methods: Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, Latent-WCT/WCT, StyleID, AdaIN, and WEAVE as applicable. R5-WikiArt contains 11 entries because not every diagnostic baseline was rerun on the R5 board. The paper table is manually aligned to these sidecars and the CLIP/LPIPS summaries.

Metric interpretation caveats. DINO-S and CLIP-S are style-reference similarities, not human preference labels. DINO-C and LPIPS quantify source preservation, but they penalize different failure modes: LPIPS is sensitive to perceptual pixel/feature distance, whereas DINO-C is a semantic feature similarity. The paper therefore treats the result as a multi-objective operating point. A method with high DINO-S and very high LPIPS is not considered a content-preserving transfer method even if it appears strong on style alone.

Main Results Used by the Paper

The complete multi-method ledgers for D5-512, P2A-256, and R5-WikiArt appear in the main paper Table 1; we do not restate those rows here. For convenience, the WEAVE board means used by the paper are:

Board	DINO-S	CLIP-S	LPIPS	DINO-C
D5-512	0.4916	0.7128	0.2593	0.8103
P2A-256	0.4801	0.6681	0.3116	0.8612
R5-WikiArt	0.5226	0.7747	0.2895	0.7717

The D5-512 claim is a balanced Pareto claim rather than a winner-on-every-axis claim: StyleAligned reaches higher raw DINO-S but with severe content loss, while WEAVE keeps low LPIPS, high DINO-C, about one million trainable parameters, and a 1.4-minute training time on one RTX 3060. The D5 point uses the oriented route; the other boards retain the base model, as stated in the main-table caption.

Paired Significance Tests on D5

The main paper reports mean metrics only. Using the same per-sample metric CSVs that underlie main-paper Table 1, we run a two-sided paired t-test on each metric (n=750 except Seedream n=720). Table S4 reports Δ = WEAVE$-$baseline; positive Δ is better for DINO-S/CLIP-S/DINO-C and worse for LPIPS. Against our SaMam retrain (CLIP-S 0.702, LPIPS

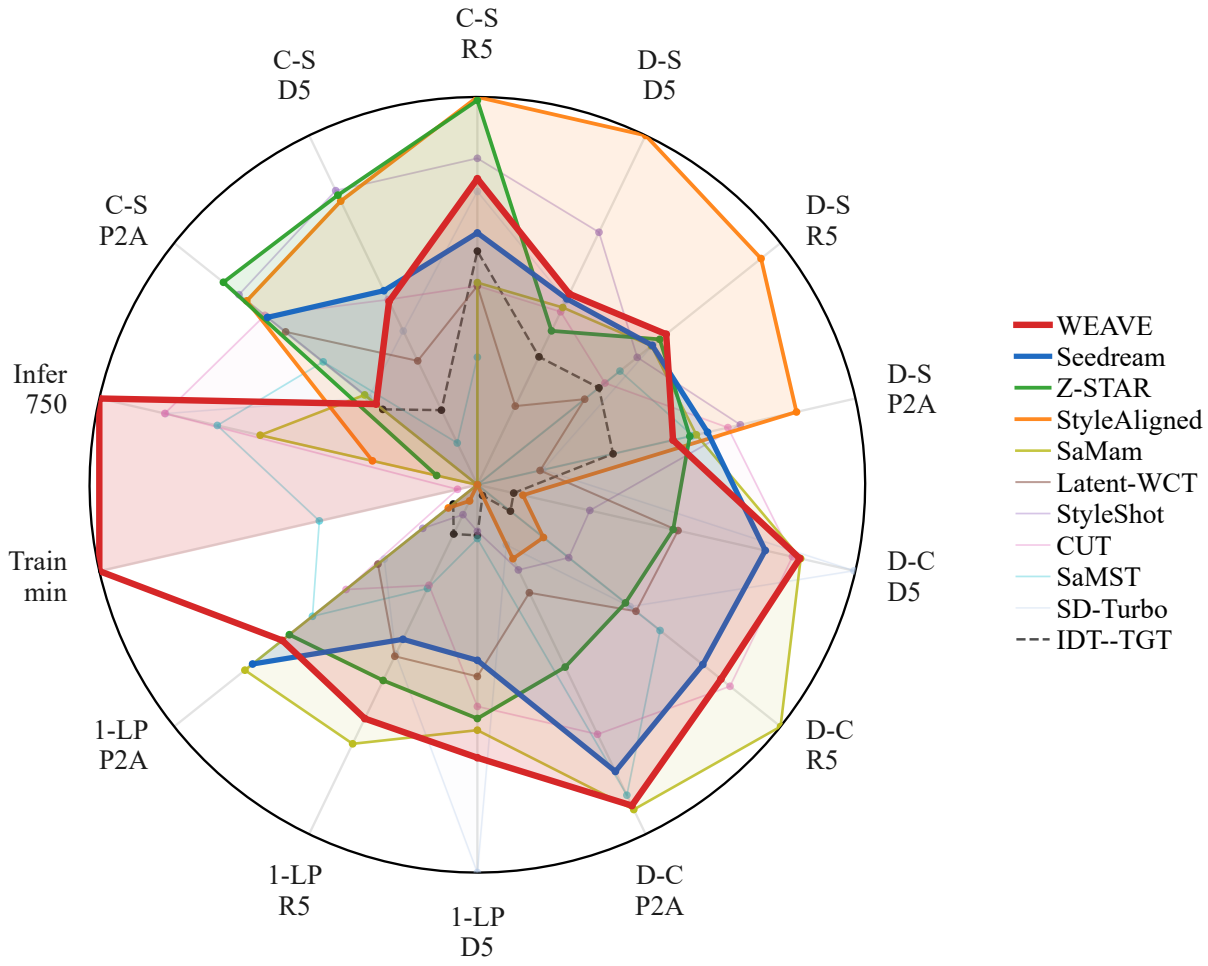


Figure S5: Style, content, and cost across three benchmarks. The dashed boundary uses IDT on style axes and TGT on content axes. Each metric family is min–max normalized once across all reported methods and datasets; speed uses the same normalization after $\log_{10}(1+t)$, where t is seconds. Raw values are those of main-paper Table 1.

0.248), WEAVE has significantly higher DINO-S and CLIP-S with statistically indistinguishable content metrics under this pairing; relative to StyTR-2, AesPA-Net, and Seedream, WEAVE trades style for content, consistent with the main operating point.

Memory-Only Style Registration

This section details the registration experiment reported in the main paper (Memory-only style registration, Generalization and Robustness).

Registration split. Registration uses the held-out WikiArt registration split: the five Other5 families (Abstract Expressionism, Art Nouveau / Modern, Cubism, Expressionism, Symbolism), each of which is held out from the D5 collection. For each family we draw 30 registration images and 30 evaluation images from disjoint pools; the 30 evaluation images per style form the same 750-pair Other5 board used for the zero-shot comparison (Table 4, main paper). The registration and evaluation image lists per family are included in the anonymous package.

Baseline	DINO-S	CLIP-S	LPIPS	DINO-C
CUT	$+0.012^{***}$	$+0.012^{***}$	-0.072^{***}	-0.006 ns
StyTR-2	-0.018^{***}	-0.004 ns	-0.359^{***}	$+0.042^{***}$
AesPA-Net	-0.037^{***}	-0.019^{***}	-0.311^{***}	$+0.188^{***}$
SaMam	$+0.014^{***}$	$+0.007^{***}$	$+0.006$ ns	-0.002 ns
Seedream 4.5	-0.008^{*}	-0.014^{***}	-0.153^{***}	$+0.049^{***}$

Table S4: Paired t-test on D5: mean difference Δ = WEAVE $-$ baseline from the same per-sample metric CSVs as main-paper Table 1 (SaMam CLIP-S 0.702, LPIPS 0.248). Positive Δ is better for DINO-S/CLIP-S/DINO-C, worse for LPIPS. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; ns: not significant. CUT/StyTR-2/AesPA-Net/SaMam use 750 pairs; Seedream uses 720 overlapping pairs.

Optimization. Starting from the trained D5 seed-42 checkpoint, we append five new memory entries (one bank of 256×384 tokens per family) and freeze all existing parameters: the velocity trunk, the LL/LH/HL routed heads,

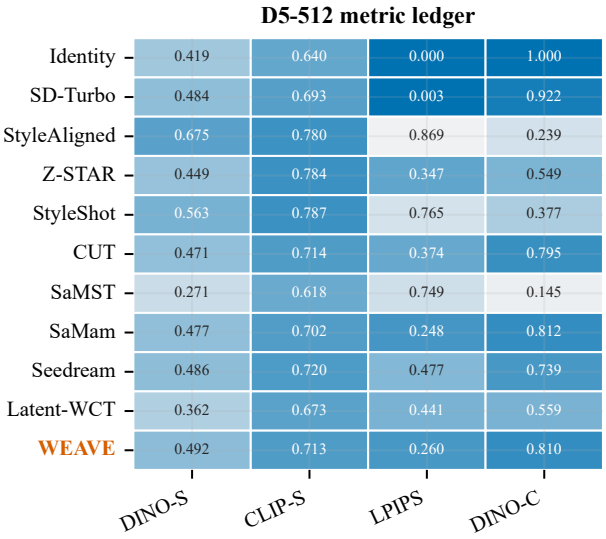


Figure S6: D5-512 metric ledger as a heatmap. LPIPS cells are color-normalized after inversion for readability, while annotations show raw values from main-paper Table 1.

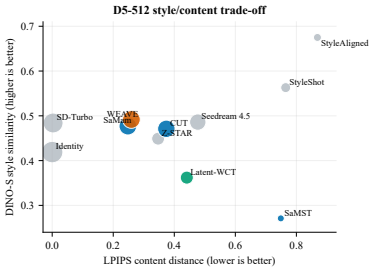


Figure S7: D5-512 style/content trade-off. Marker size grows with DINO-C. WEAVE is not the maximum-style point, but it occupies the low-LPIPS, high-content region among learned methods that clear the identity style floor.

the frozen VAE, the analytic operator, and the five existing D5 memory entries. New entries are initialized with $0.02 \cdot \mathcal{N}(0, I)$, matching the style-memory initialization scale used during D5 training. Optimization uses AdamW (learning rate 2×10^{-4}, weight decay 10^{-4}) with the same flow-matching objective as training, for a fixed 3,000-step budget with no evaluation-metric checkpoint selection. Memory-only registration uses batch size 4 and optimizes only the appended entries (0.098M parameters per family, 0.49M total); full-field fine-tuning uses batch size 2 and optimizes all 1.53M parameters of the expanded model. Both variants start from the same appended-memory initialization and use identical registration data and the same inference configuration (β=2.0, η=0.1, K=8). Stochasticity is limited to the memory initialization and minibatch sampling.

Timing. Memory-only registration of all five families takes approximately 98 s (3,000 steps, batch 4); full-field fine-tuning takes approximately 4 min (3,000 steps, batch 2). Both are measured on the same single RTX 3060 (12 GB) used for D5 training; the timing excludes the one-time offline VAE encode of the registration images. Evaluation reuses the same DINOv2-small protocol as Table 4.

Target family	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑
Abstract Expressionism	0.546	0.798	0.264	0.822
Art Nouveau / Modern	0.551	0.763	0.261	0.810
Cubism	0.557	0.786	0.275	0.792
Expressionism	0.573	0.788	0.256	0.832
Symbolism	0.584	0.784	0.268	0.804
All (750 pairs)	0.562	0.784	0.265	0.812

Table S5: Per-target-family breakdown of memory-only registration on the 750-pair Other5 board (150 pairs per family, including the same-style diagonal). Every family clears the aggregate zero-shot DINO-S of 0.531.

Per-style breakdown. Table S5 reports memory-only registration metrics per target family (150 pairs each). All five families exceed the aggregate zero-shot DINO-S of 0.531, so the overall gain is not driven by a single style.

No forgetting on registered D5 styles. Registration freezes the D5 trunk, routed heads, analytic operator, and the five existing D5 memory entries, and appends new entries at higher indices; the D5 forward path therefore does not read the new entries. Consequently, registering Other5 leaves the D5 outputs unchanged up to numerical determinism, and no re-evaluation of the D5 board is required to establish no-forgetting.

Registration image-count curve. To probe the data requirement of registration, we vary the number of registration images per family $n \in \{1, 5, 10, 30\}$ (nested subsets of the 30-image pool) while keeping the fixed 3,000-step memory-only budget and all other settings, and evaluate on the same 750-pair board. Quality is already saturated at a single reference image: DINO-S rises from 0.558 at n=1 to 0.562 at n=30 (a gain of only 0.004), and CLIP-S, LPIPS, and DINO-C stay within their n=30 values throughout (Table S6). Registration wall-clock is essentially independent of n because the 3,000-step budget dominates; on our hardware it varies only between 60 s and 105 s across n=1 . . . 30 (the RTX 3060 timing above reports ≈98 s for n=30). Together these show that registering a new style is a low-burden operation: supplying more reference images barely changes the cost yet yields negligible further gain beyond the first image.

Hyperparameter Sensitivity

This section reports one-factor sweeps of the solver-step count K and the inference-time analytic alignment strength β and style-extrapolation η on the D5-750 board. The headline finding is that WEAVE is stable over $K \in \{2, 4, 8, 16\}$ and $\beta \in [1, 2]$ (CLIP-S varies by < 0.001 across K; the style–content trade-off is preserved throughout $\beta \in [1, 2]$), while $\beta \geq 2.5$ collapses content and larger η shifts the operating point toward content preservation. β=2.0, η=0.1, K=8 is the reported operating point; no D5-test-board search reselects it.

n	DINO-S↑ (all/off)	CLIP-S↑	LPIPS↓	DINO-C↑	Time (s)
1	0.558 / 0.492	0.782	0.260	0.820	60
5	0.561 / 0.498	0.784	0.269	0.812	101
10	0.561 / 0.498	0.784	0.270	0.808	102
30	0.562 / 0.499	0.784	0.266	0.813	105

Table S6: Registration image-count curve on the 750-pair Other5 board. Each row registers the five Other5 families from the D5 checkpoint using n registration images per family (nested subsets of the 30-image pool) and a fixed 3,000-step memory-only budget. DINO-S is reported over all 750 pairs and the 600 off-diagonal pairs. Quality is saturated at a single reference image, and registration time is essentially independent of n.

α	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑
0.1	0.4842	0.6956	0.2736	0.8322
0.2	0.4843	0.6979	0.2630	0.8291
0.3 (default)	0.4916	0.7128	0.2593	0.8103
0.4	0.4871	0.7029	0.2570	0.8226
0.5	0.4871	0.7049	0.2582	0.8170

Table S7: Endpoint LL-alignment strength α on D5-750 (one-factor retrain).

Unless noted, sweeps use the D5-750 board and seed 42. Training-time parameters (α, λ_{LL}) retrain with the same evaluation-free stopping rule when applicable. Inference-time parameters reuse a fixed checkpoint and vary only the listed knob. In the solver implementation, HF $\leftarrow (1-\beta)\text{HF} + \beta\,\mathcal{A}(\text{HF}; (1+\eta)q_s)$ with $\beta =$ `endpoint_adain_scale` and $\eta =$ `style_extrap_alpha`. Training defaults are α=0.3, λ_{LL}=0.3, and σ=0.02. Main-table evaluation uses β=2.0, η=0.1, K=8 via `inference.json` (β is solver-only and does not enter the training loss). One-factor diagnostic sweeps below (including other β settings) illustrate trade-offs and stability boundaries; where a default row does not exactly match the oriented main-table metrics, the trend—not the absolute score—is the intended takeaway, and no D5-test-board search reselects the main-row config.

Source-alignment strength α. Table S7 and Figure S8 vary $\tilde{\ell}_s = (1-\alpha)\ell_c + \alpha\,\text{AdaIN}(\ell_c, \ell_s)$. The method remains stable over $\alpha \in [0.2, 0.5]$, with α=0.3 providing the strongest style affinity among the scanned settings while retaining low LPIPS.

LL loss weight λ_{LL}. Main-paper Table 2 already compares λ_{LL}=0.3 and 1.0 (final metrics nearly unchanged). The matched gradient probe in the Information-Flow Diagnosis section shows that λ_{LL}=1.0 drives the LL residual/gradient near zero while HF loss is essentially unchanged, whereas 0.3 keeps both routes active. Thus LL weighting is primarily a responsibility-balancing mechanism rather than a sharply tuned quality parameter: the final operating point is robust over a broad nonzero range, while the weight prevents early LL saturation of the gradient.

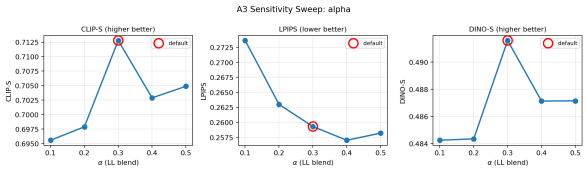


Figure S8: Style–content variation under α (default in Table S7).

β	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑
1.0	0.4832	0.7055	0.2783	0.8024
1.5	0.4842	0.7061	0.2787	0.8030
2.0	0.4916	0.7046	0.2548	0.8105
2.5	0.3140	0.6792	0.5630	0.2738
3.0	0.2714	0.6590	0.7027	0.2076

Table S8: Inference-time β=`endpoint_adain_scale` on a fixed diagnostic checkpoint (qualitative trends; absolute CLIP-S/LPIPS need not match the oriented main row).

Analytic alignment strength β and η. Tables S8–S9 and Figures S9–S10 vary the fiber update at inference on a fixed diagnostic checkpoint. The solver is stable over the tested range $\beta \in [1, 2]$, while larger extrapolation crosses a clear stability boundary between 2.0 and 2.5 (content collapses for $\beta \geq 2.5$). Increasing η under the current affine update shifts the operating point toward stronger content preservation, while smaller η yields higher style affinity; η=0.1 is the main-config default.

Solver steps K. Table S10 varies only `num_steps` $\in \{2, 4, 8, 16\}$ on a fixed D5 checkpoint. CLIP-S changes by less than 0.001; LPIPS improves from K=2 to K=4 and then plateaus; relative generation time grows nearly linearly with K. Absolute end-to-end generation for the oriented main model is 73 s/750 (97.3 ms/image; ONNX/TRT encode–decode + ODE, batch settings in the timing protocol above); the "Rel. ODE" column here is a relative kernel-time probe and is not the same wall-clock protocol. We use K=8 as a conservative default after quality largely saturates by K=4.

Endpoint AdaIN scale (solver evaluation). β scales the analytic fiber update S_s at inference and does not enter the training loss. The base checkpoint configuration stores β=1.0. Main Table 1 uses the fixed override β=2.0 in `inference.json` (verified against the D5 board means 0.4916/0.7128/0.2593/0.8103).

Baseline Accounting

External baselines are separated into three accounting classes:

- **Training-free diffusion/editing baselines:** SD-Turbo, StyleAligned, Z-STAR, StyleShot, Seedream 4.5. Their large frozen backbones are reported with superscript parameter counts in the main table; training is marked as free or API.
- **Learned style-transfer baselines:** CUT, SaMST, SaMam. For the D5 main table, SaMam is retrained by us on the

η	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑
0.0	0.5016	0.7127	0.2730	0.7650
0.05	0.4967	0.7095	0.2635	0.7890
0.1	0.4916	0.7046	0.2548	0.8104
0.2	0.4862	0.6968	0.2436	0.8369

Table S9: Inference-time η=`style_extrap_alpha` on the same diagnostic checkpoint as Table S8.

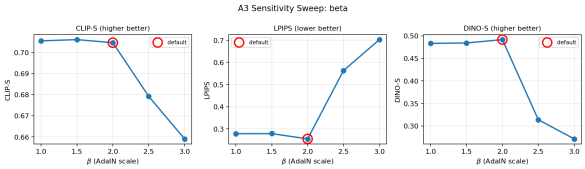


Figure S9: Style–content variation under β (Table S8).

same five-style training split used by WEAVE (not a generic arbitrary-style checkpoint), and train/infer times are measured on RTX 3060-class hardware when available.

- **Analytic wavelet/statistics baseline:** Latent-WCT. This has zero trained parameters and is included to show that wavelet statistics alone do not explain the WEAVE result.

All methods are evaluated on the same ordered board where outputs are available. Missing or withheld speed/parameter values are marked as unavailable rather than imputed.

Baseline Data Inventory

The baseline ledger contains both measured baselines and externally sourced or API-limited methods. The evaluation convention is:

1. collect generated images under a method/board/run directory;
2. normalize filenames to the ordered source-target board;
3. evaluate CLIP-S, LPIPS, and optional side metrics using the shared evaluator;
4. evaluate DINO-S and DINO-C using the DINO sidecar scripts;
5. record unavailable timings explicitly rather than estimating them from unrelated hardware.

For D5-512, the paper table includes Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, Latent-WCT, and WEAVE. For P2A-256 and R5-WikiArt, the same naming convention is used, but some entries have fewer available images or omitted timing fields. This is why each sidecar records image counts and skipped counts rather than assuming all methods have a complete board.

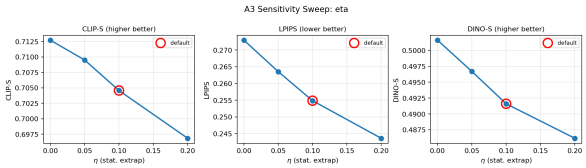


Figure S10: Style–content variation under η (Table S9).

K	CLIP-S ↑	LPIPS ↓	Rel. ODE (s)	Wall+metrics (s)
2	0.7164	0.2921	4.6	60.3
4	0.7162	0.2875	8.9	55.2
8	0.7167	0.2870	18.0	64.6
16	0.7171	0.2874	36.0	83.1

Table S10: Solver-step sensitivity on D5-750 (relative ODE probe; main-paper end-to-end generation is 73 s at K=8 under the ONNX/TRT pipeline).

Method group	Output source	Notes
Identity	evaluator no-op	Generated by copying/reading source image as output; defines style floor and content ceiling.
Latent-WCT	analytic transform	Zero trained parameters; evaluates whether wavelet statistics alone are sufficient.
CUT / SaMST / SaMam	learned baselines	SaMam is retrained on the same five-style D5 split as WEAVE and evaluated on the same 750 pairs; train/infer times measured on RTX 3060-class hardware when reported.
SD-Turbo / StyleAligned / Z-STAR / StyleShot	frozen-backbone editing baselines	Large pretrained backbones; no task-specific training counted, but inference wall-clock is reported when outputs were generated in the evaluation run.
Seedream 4.5	API outputs	Parameter count and internal runtime are withheld; metrics are computed from saved outputs on the shared board.
WEAVE	single-RTX-3060 run	Main checkpoint and timing records come from the audited experiment tree.

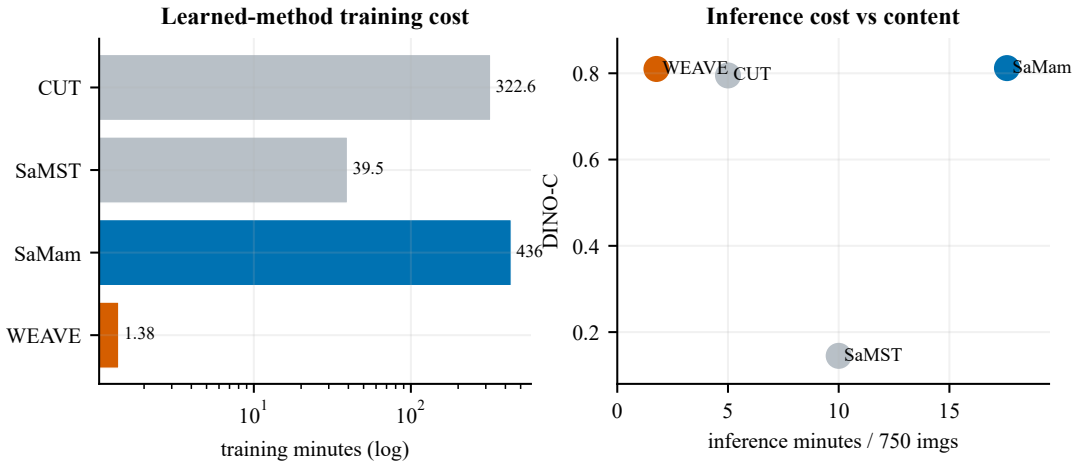


Figure S11: Cost accounting for learned baselines. WEAVE trains in minutes rather than hundreds of minutes and remains in the high-content region at inference.

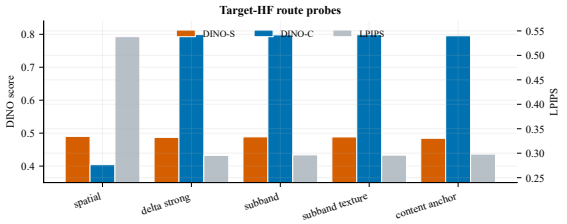


Figure S12: Target-HF route ablation. Spatial target-HF injection raises style but substantially reduces content scores, while pooled/subband target-HF routes preserve content at nearly the same style level.

Ablation Summary

The ablations support three complementary responsibilities: source-aligned endpoint construction, learned LL/LH/HL residual transport, and stepwise analytic HF alignment. Learned style memory provides a coarse registered-style prior rather than the main image-specific style injector.

The cross-attention branch is treated as auxiliary. Probe runs show near-closed gates around 0.056–0.061 and weak gradient mass on the style-memory branch, yet the no-memory subband run underperforms the standard subband route. The effective style movement comes mainly from stepwise AdaIN/statistics, flow transport, and coordinate-free target-HF codes; style memory supplies a useful coarse category prior but cannot replace the image-specific HF route.

Analytic-alignment operator variants. The main model uses a single-level Haar decomposition and a global (channel-wise) high-frequency alignment. We test two inference-time operator variants that keep the trained field fixed: a *two-level Haar* alignment (locks the coarser LL_2 and aligns both Haar levels) and a *block-local* HF alignment (per-16 $\times$ 16 spatial-window mean/std matching instead of global). Both are applied at inference only, so they add no parameters and negligible generation overhead. On the D5-750 board they land within the baseline operating point: two-level Haar shifts slightly toward content (DINO-C 0.819 vs. 0.811, LPIPS 0.256 vs. 0.259) at a small DINO-S cost, while block-local HF raises CLIP-S (0.718 vs. 0.713) but lowers DINO-C (0.788). Neither improves the balanced point, supporting the single-level global design.

Variant	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	Time (7
Single-level, global (main)	0.4919	0.7128	0.2593	0.8107	
Two-level Haar	0.4891	0.7110	0.2559	0.8186	
Block-local HF (16^2)	0.4901	0.7185	0.2695	0.7878	

Table S11: Inference-time analytic-alignment operator variants on D5-750 (frozen D5 checkpoint, β=2.0, η=0.1, K=8). All are parameter-free operator changes; generation time is per 750 pairs in PyTorch (the 73 s/750 main figure uses ONNX/TRT).

Corrected Controls and Stability Checks

The inference-time AdaIN control was rerun from the reported epoch-4 checkpoint after verifying that the active nested configuration field was set to zero. All 20 checked corresponding PNG hashes differ from the full model. Removing AdaIN changes DINO-S from 0.4916 to 0.4821 and CLIP-S from 0.7128 to 0.7005, while DINO-C increases from 0.8103 to 0.8455 and LPIPS decreases from 0.2593 to 0.2258. Thus the operator is active and induces the intended style–content trade-off; the earlier no-effect result is not used.

For the matched training control, changing λ_{LL} from 0.3 to 1.0 changes peak DINO-S only from 0.4917 to 0.4910. The 1.0 run reaches its peak at epoch 3 and the frozen probe selects epoch 5 with regret 6.8×10^{-5}. This supports robustness to a broad nonzero LL weight, not a claim that 0.3 is uniquely optimal. In contrast, removing the source-aligned endpoint yields peak DINO-S 0.4894 with LPIPS 0.3138, and the

probe fires eight epochs before that peak; this is the stronger endpoint-specific ablation.

Finally, reference-pool stability is evaluated without regenerating images. For each of 1,000 repetitions and each target style, one size-m subset is sampled without replacement from its 30 held-out references and reused for all 150 requests to that style. The board mean of DINO-S(WEAVE) minus DINO-S(IDT) is positive in all repetitions: mean 0.0298 with 95% interval [0.0204, 0.0394] for $m = 8$, and mean 0.0319 with [0.0225, 0.0380] for $m = 16$. The corresponding per-request positive fractions are about 69%, so this is evidence for a stable board-level conclusion, not an assertion that every source improves.

Alternative VAE Portability

To test whether WEAVE depends on the SD1.5 EMA latent space, we replace its frozen VAE with FLUX.1, FLUX.2, and Qwen-Image and retrain the wavelet-routed model. Only the latent-channel projections are adapted to each VAE; the D5 data, one-level Haar routing, and evaluation board are retained. This cross-VAE experiment uses the learned HH variant, whose style–content trade-off is analyzed below.

The three VAEs all yield comparable style-transfer results, showing that WEAVE does not depend on the older SD1.5 EMA representation and remains effective with newer VAE families. The FLUX.2 run reaches the same DINO-S as the reported SD1.5 result while improving CLIP-S, LPIPS, and DINO-C. Their optimization trajectories nevertheless differ: under the fixed internal rule the selected epochs are 6 (FLUX.1), 29 (FLUX.2), and 1 (Qwen-Image). Frequency routing transfers across tokenizers; the absolute gate/gradient threshold still shifts with latent geometry, so cross-tokenizer early stopping benefits from per-tokenizer trajectory normalization.

HH trade-off with a high-capacity VAE. On FLUX.2, increasing the HH velocity-loss weight from 0 to 20 changes DINO-S from 0.4911 to 0.4870 and CLIP-S from 0.6804 to 0.6740, while DINO-C rises from 0.8763 to 0.8888 and LPIPS falls from 0.2359 to 0.2229. The intermediate weights follow the same trend. Learning HH therefore provides a smooth content–style trade-off rather than a uniformly beneficial or harmful route, consistent with retaining the simpler statistical HH treatment in the main configuration.

Frozen SD1.5 Extension

We test whether the wavelet alignment operator transfers independently of the compact velocity model. The control is a frozen SD1.5 image-to-image pipeline with an empty prompt, strength 0.5, and 20 DDIM steps. The extension applies one-level Haar decomposition to the output latent, aligns the channel-wise moments of LH, HL, and HH to the target-style latent at scale 0.8, leaves LL unchanged, and then applies the inverse transform. No SD1.5 weights are updated. On the 600 cross-style D5 pairs, the base and extended editors obtain respectively DINO-S 0.3974/0.4080, CLIP-S 0.7221/0.7366, LPIPS 0.3591/0.3630, and DINO-C 0.6589/0.6531. Both style gains are significant under paired t-tests ($p < 10^{-15}$). On all 750 pairs, DINO-S changes from 0.4525 to 0.4613 and CLIP-S from 0.7476 to 0.7594, while LPIPS remains 0.3586/0.3588 and DINO-C changes from 0.6583 to 0.6539. Timing over 720 completed requests gives 4.9 ms per image for the DWT–AdaIN–IDWT operation. The extension therefore isolates the portability of the frequency-domain operator; it is not used to equate the compact model with SD1.5.

Reproduction Commands

Commands assume the minimal code package root as the working directory. Data and ONNX graphs must be prepared locally (Section S2).

Listing 1: Environment setup

```
python -m venv .venv
# activate .venv (Windows: .venv\Scripts\activate)
pip install torch torchvision --index-url https://
    download.pytorch.org/whl/cu128
pip install -r requirements.txt
# optional: pip install -r requirements-dev.txt #
    pytest
```

Listing 2: Train the D5-512 paper checkpoint

```
# requires local data/train latents
python run.py --config configs/
    hf_oriented_internal_early_stop.json
```

Listing 3: Evaluate the selected D5 checkpoint (Table 1 D5 row)

```
python -m utils.run_evaluation \
  --checkpoint checkpoints/weave_epoch_0004.pt \
  --test_dir data/test \
  --output eval_outputs/d5_750_rerun \
  --config config.json \
  --config_override inference.json \
  --num_steps 8
```

Listing 4: Zero-shot diagnostic on R5 using the D5 checkpoint

```
# Does NOT reproduce the collection-specific R5 row in
     main-paper Table 1.
python -m utils.run_evaluation \
  --checkpoint checkpoints/weave_epoch_0004.pt \
  --test_dir data/images/r5/test \
  --output eval_outputs/r5_d5ckpt_zeroshot \
  --config config.json \
  --config_override inference.json \
  --num_steps 8
```

Listing 5: Generation-only timing (PyTorch fallback)

```
# Matches metric-free wall-clock protocol; the
    reported 73\,s figure uses ONNX/TRT
# (Section~\ref{sec:supp-onnx}), not this pure-PyTorch
     path.
python -m utils.run_evaluation \
  --checkpoint checkpoints/weave_epoch_0004.pt \
  --test_dir data/test \
  --output eval_outputs/timing \
  --config config.json \
```

Frozen VAE	Latent channels	Reported epoch	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑
FLUX.1	16	6	0.4889	0.7134	0.2556	0.7801
FLUX.2	32	29	**0.4918**	**0.7165**	**0.2388**	**0.8728**
Qwen-Image	16	1	0.4775	0.7144	0.2631	0.8251

Table S12: Results after replacing the SD1.5 EMA VAE with recent frozen VAEs on D5. Each row uses the same evaluation-free internal-dynamics selection protocol as the main SD1.5 run (reported epoch is the selected training step under that rule).

```
--config_override inference.json \
--clip_backend none \
--eval_disable_lpips \
--num_steps 8 \
--profile_timing \
--force_regen
```

End-to-End Checklist

1. Install the Python environment and confirm CUDA sees one RTX 3060-class GPU.
2. Prepare local `data/train` and `data/test` (not in the minimal archive).
3. Train with `configs/hf_oriented_internal_early_stop.json`, or use `checkpoints/weave_epoch_0004.pt`.
4. Evaluate D5 with `inference.json` (β=2.0).
5. Optionally export ONNX and compile TensorRT to match the 73.0 s timing ledger.
6. Treat R5/P2A Table 1 rows as collection-specific models not shipped in the minimal archive.

ONNX Export and TensorRT Compilation

The reported 73.0 s / 750 generation-only figure uses an optional accelerated VAE path. **ONNX graphs and TensorRT engines are not included in the minimal anonymous archive** because they exceed the package size budget ($\approx$65–99 MB per graph). They can be rebuilt from the public SD1.5 EMA VAE as follows.

Export. Encoder and decoder are exported separately with fixed batch size 8 (FP16) from the public SD1.5 EMA VAE. WEAVE itself remains a PyTorch field; only the frozen VAE is exported. The decoder latent scale `0.18215` is applied outside the ONNX graph. The minimal archive does not include export utilities or ONNX binaries; the audited settings and stage-level timing are documented in `docs/onnx_export_and_trt.md` for rebuilds.

TensorRT compile. Recommended: batch 8, FP16, workspace 2 GB. On RTX 3060 12 GB this compiles the decoder in about 15 minutes and yields $\approx$ 78.6 ms/image decode. Batch 16 requests $\approx$10.7 GB during build and OOMs on 12 GB cards (engine falls back to weaker tactics and can be slower). Workspace 4 GB improves decode by only $\approx$0.6%.

Pipeline timing (audited). With batch 8 and TRT workspace 2 GB on one RTX 3060 12 GB: encoder+WEAVE 14.0 s (18.7 ms/image), TRT decoder 59.0 s (78.6 ms/image), total **73.0 s** (97.3 ms/image; 10.3 pairs/s). Ledger: `docs/generation_only_timing_summary.json`.

Sanitized Artifact Ledger

Known Limits

The reported evidence supports a multi-objective operating point under automatic metrics rather than universal dominance across metrics or datasets. The alternative-VAE experiments establish compatibility with several recent latent representations, but do not establish identical optimization dynamics, compute, or optimal stopping across tokenizers.

Table S13: Artifacts for reproduction. Minimal archive contents vs. external requirements.

Artifact	In archive?	Notes
`configs/hf_oriented_internal_early_stop.json`	yes	D5 train config.
`config.json` / `inference.json`	yes	Base config; eval β=2.0 override.
`checkpoints/weave_epoch_0004.pt`	yes	Selected D5 checkpoint only.
`tools/prepare_submission_data.py`	yes	Path sanitize for local data.
`docs/generation_only_timing_summary.json`	yes	73.0 s timing ledger.
`docs/onnx_export_and_trt.md`	yes	ONNX/TRT rebuild protocol.
`data/README.md`	yes	Expected local data layout.
`data/train`, `data/test`	no	Prepare from WikiArt + VAE encode.
`data/onnx_exported/`	no	Export; not shipped (size).
P2A/R5 collection-specific checkpoints	no	Not required for D5 Table 1 row.